Mehmet Çalıkoğlu

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Education

Dokuz Eylul University — Izmir, Turkey
BSc in Computer Engineering — *Aug 2023 – Aug 2026*

Experience

Game Developer Intern — Jul 2026 – Aug 2026
Furtle Game — *Izmir, Turkey*

- Developed gameplay features for a Unity mobile game using C# and object-oriented programming, collaborating with a team of designers and developers
- Built a playable prototype in Unity by implementing core gameplay mechanics, event-driven game state system, and UI interactions that resulted in a fully functional demo delivered within 4 weeks
- Applied MVC architecture to structure gameplay logic, presentation, and user input that improved code maintainability and readability

Software Engineer Intern — Aug 2024 – Sep 2024
Optimimax — *Izmir, Turkey*

- Developed a Unity-based digital twin application to visualize real-time factory operations, translating live sensor and machine data into a 3D simulation
- Integrated REST API endpoints to stream live production data with low latency, allowing users to monitor factory metrics in real time
- Designed the software architecture using SOLID principles to improve modularity and maintainability

Projects

Smart Greenhouse Monitoring System | *C++, Python, Flutter, PostgreSQL, Docker* — Oct 2025 – Jun 2026

- Engineered embedded firmware for an ESP32 microcontroller in C++, implementing a finite-state-machine architecture across concurrent FreeRTOS tasks to manage telemetry, connectivity, and fault states
- Applied Test-Driven Development (TDD) within the firmware layer to validate sensor behavior
- Used the MQTT protocol for real-time transmission of temperature, humidity, and CO_2 readings from the microcontroller to the backend
- Built a machine learning system using Random Forest Regressor to predict crop yield from sensor data, achieving an R^2 score of 0.82

Scopapilia | *C#, Unity, Git* — Jun 2025

- Developed a 2.5D horror game in 48 hours during a game jam, which won the 3rd place out of 20 teams
- Designed and implemented core gameplay systems including player movement and audio interactions
- Collaborated in a small team of four, using Git for version control to rapidly prototype and deliver an enjoyable gameplay experience

Bitty Witch | *C#, Unity, Git* — Jan 2025

- Developed a 2.5D crafting game featuring resource collection and potion mechanics
- Implemented crafting logic, dialogue systems, and multiple interactive minigames
- Collaborated with a team of five to design and deliver a polished gameplay experience within 48 hours

Pizza Delivery 3D | *C#, Unity* — Dec 2022 – Feb 2023

- Developed a hypercasual 3D game centered on item collection and obstacle avoidance mechanics
- Implemented core gameplay loop including player input handling, collision detection, and scoring systems
- Designed prototype levels with increasing difficulty to demonstrate gameplay capabilities

Technical Skills

Languages: C/C++, C#, Rust, Python, JavaScript
Frameworks: FastAPI, .NET, React, Flutter
Game Engines: Unity, Unreal Engine, Godot
Developer Tools: Git, Docker, Linux, PostgreSQL